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(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

4th Semester, April, 2024

Paper Code: BS-202 **Subject: Probability, Statistics and Linear Programming**
Time: 1½Hrs. **Max. Marks: 30**

Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining.

Q.No.	Questions	Max. Marks	CO(s)
1(a)	Two bolts are drawn from a box containing 6 good and 8 defective bolts. Find the probability that the second bolt is good if the first one is found to be defective.	2.5	CO1
1(b)	Find the value of k and mean value of X. If X is continuous random variable with probability density function given by $$\left.\begin{aligned} f(x) &= kx \ (0 \le x \le 2) \\ &= 2k \ (2 \le x \le 4) \end{aligned}\right\}$$	2.5	CO1
1(c)	If X and Y are jointly distributed random variables, then prove that $$Var(X+Y) = Var(X) + Var(Y) + 2Cov(X,Y)$$	2.5	CO2
1(d)	What is the Central Limit theorem & why it is useful?	2.5	CO2
2(a)	A variate X has the probability distribution function x: -3 6 0 P(x): 1/6 1/2 1/3 Then find the expectation E(x) and E(x^2), hence evaluate E$(2x+1)^2$.	5.0	CO1
2(b)	In a sample of 1000 cases, the mean of a certain test is 14 and deviation is 2.5. Assuming the distribution to be normal, find (i) How many students score between 12 and 15? (ii) How many score below 8? (iii) How many score above 16?	5.0	CO1
3(a)	The join Probability distribution of two random variables X and Y is given by $P(X=0,Y=1)=1/3$, $P(X=1,Y=-1)=1/3$ and $P(X=1,Y=1)=1/3$ Find (i) the marginal distributions of X and Y. (ii) the conditional probabilities distribution of X, given by $Y=1$.	5.0	CO2
3(b)	Find the moment generating function of the exponential distribution $f(x) = \frac{1}{c} e^{-x/c}$, $0 \le x \le \infty$, $t > 0$, hence find its mean and Standard deviation if $\lvert t \rvert < \frac{1}{c}$.	5.0	CO2
4(a)	An insurance company found that only 0.01% of the population is involved in a certain type of accident each year. If its 1000 policy holders were randomly selected from the population, what is the probability that not more than two of its clients are involved in such an accident next year? (given that $e^{-0.1} = 0.9048$)	5.0	CO1
4(b)	X is the variate with mean 30 and Standard deviation 5, then find the Probability for (i) $26 \le X \le 40$ (ii) $\lvert X - 30 \rvert > 5$	5.0	CO2

Apna Engineering Wallah